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Cosmic Expansion History from SNe Ia via Geometric Variational Inference - the charm2 code

Iason Saganas[1,2], Matteo Guardiani[1,2], Natalia Porqueres[3] and Torsten Enßlin[2]

[1] Ludwig-Maximilians-University Munich, Scheinerstr. 1, 81679 Munich, Germany
e-mail: I.Saganas@campus.lmu.de
[2] Max Planck Institute for Astrophysics, Karl-Schwarzschildstr. 1, 85748 Garching, Germany.
[3] Université Paris-Saclay, Université Paris Cité, CEA, CNRS, AIM, 91191, Gif-sur-Yvette, France

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ABSTRACT

Context. Cosmological analyses using the latest set of type Ia Supernova data favour an evolving dark energy model, challenging the standard ΛCDM cosmological paradigm. Nonparametric reconstructions of the expansion history help examine the assumptions we impose on the data by, e.g., uncovering signal features that a parametric model might miss. Information Field Theory (IFT) is a Bayesian framework for optimal, nonparametric reconstruction algorithms.
Aims. In this work, we present charm2, the successor to charm1, a previous IFT-based code to reconstruct the cosmic energy density's redshift evolution from Supernovae Ia. We employ a model that strongly mitigates prior cosmological biases. We apply our reconstruction algorithm to the Union2.1, Pantheon+, DESY5 and DESY5-Dovekie data sets to investigate the agreement between the nonparametric reconstruction and the signal suggested by a parametric, flat ΛCDM model.
Methods. We use the geometric variational inference algorithm which finds a coordinate transformation through which a potentially highly curved posterior distribution can be "flattened out" and thus be well approximated by a Gaussian distribution. We trace the redshift evolution of the energy density on a double-logarithmic scale, which, after de-trending, can be well described by a stationary stochastic process. The stationary process is part of a flexible Bayesian hierarchical model that enables inference of central cosmological and statistical aspects of the signal. A slight reformulation of the instrument response compared to the charm1 predecessor removes the need to assume a fiducial value for H_0 in the inference.
Results. The reconstructions using the Union2.1 and Pantheon+ data sets are consistent with parametric flat ΛCDM signal fields, although in both cases the reconstruction uncertainties are larger than those of their charm1 counterparts. Our DESY5 reconstruction deviates significantly from flat ΛCDM comparison fields and is instead compatible with an evolving dark energy signal, even for the updated DESY5-Dovekie data set. Previous studies report the presence of a systematic offset in DESY5 distance moduli at low redshift, potentially mimicking evolving dark energy. Yet, introducing an offset of the suggested magnitude in this regime does not reproduce a comparable feature in our synthetic reconstructions. Using the evidence lower bound measure for model-selection, we find no conclusive evidence for non-flat-ΛCDM features in any of the data sets. However, we note that evidence lower bound and by extension the evidence, might be a suboptimal model selection statistic for supernovae Ia alone at current noise levels.

Key words. Cosmic Energy Density – Nonparametric Bayesian Inference – Information Field Theory – Supernovae Ia

1. Introduction

The detection of the universe's accelerated expansion by Perlmutter et al. (1999) implied that most of the universe's energy content is due to an unknown energy density, called dark energy, which appears to remain constant even as the universe expands. Together with observations providing strong evidence for the existence of a cold dark matter component (such as the weak lensing map of the bullet cluster, rotation curves of galaxies, etc, see Freese (2009) for a review), the so-called ΛCDM model has been the favoured candidate for a standard model of cosmology. Cosmological analyses based on this model revealed that only about 5% of the universe's energy content is of baryonic nature, with approximately 95% bound up in the dark matter and dark energy components (Freese 2009).

In recent years, models that allow the dark energy density, ρ_Λ, to evolve over cosmic time have regained traction, with recent analyses of the Dark Energy Spectroscopic Instrument (DESI) favouring a dynamical dark energy scenario (Adame et al. 2025). Often, a parametric form is assumed for the evolution of the dark energy component. Cosmological parameters, such as the relative energy density fractions in the universe, can then be extracted by minimizing the χ^2 between the predicted and observed data with respect to the parameters of interest. If the assumed model, whether the ΛCDM model or a parametric evolution of dark energy, is incorrect, the analysis results might be biased.

To circumvent such model bias, different methods have been developed to extract cosmological information from observed data without assuming a physical model, and/or without relying on a parametric model. In the former case, we speak of an "agnostic inference"; in the latter, of a "nonparametric inference". Usual nonparametric reconstruction methods are Gaussian Processes (Holsclaw et al. 2010, 2011; Shafieloo et al. 2012; Seikel et al. 2012; Jiang et al. 2024; Mukherjee & Sen 2024), Principal Component Analyses (PCA) (Ishida & de Souza 2010; Sharma et al. 2022), Splines and other polynomial-based approaches (for example Padé-Polynomials) (Joy et al. 2019;

Benetti & Capozziello 2019; Jiang et al. 2024), as well as smoothing approaches (Shafieloo et al. 2006; Matthewson & Shafieloo 2024). In recent years, Neural-Network-based reconstructions have emerged as well (Mukherjee et al. 2022; Zhang et al. 2024).

Porqueres et al. (2017) proposed a nonparametric and agnostic approach to reconstruct the evolution of the cosmic expansion in the late universe through Supernovae Ia (SNIa) inside the framework of information field theory (IFT) (Enßlin et al. 2009). That approach relied on a Wiener filter to obtain the posterior distribution and a linearised response operator, in which the solution is perturbed and the Wiener filter is applied recursively until a stationary solution is reached. This describes an iterative maximum a posteriori scheme in which the posterior mode is approximated by a Gaussian.

Here, we improve this approach by using the geometric variational inference algorithm (`geoVI`, Frank et al. 2021) to account for the nonlinear terms of the data model and to provide more accurate posterior uncertainties.

Rather than defining constant piecewise functions (Zheng & Li 2017) or polynomials (Sharma et al. 2022), our method does not rely on any functional basis and further accounts for the full Fisher information metric, as opposed to PCA approaches which often truncate the eigenvalues of the Fisher information metric. As in neural network frameworks (e.g., Zhang et al. (2024)), our approach introduces hyperparameters but does not require training datasets.

In this paper, we present `charm2` (cosmic history agnostic reconstruction method No. 2), a code for nonparametrically reconstructing the redshift evolution of the universe's energy density, with improved treatment of nonlinearities and uncertainties when compared to its predecessor in Porqueres et al. (2017). `charm2` is built using the Python package `NIFTy` (v8[1]).

The paper is structured as follows: In Section. 2, we introduce the concepts necessary to set up our agnostic inference approach of the cosmic expansion history, defining the inferred signal and the data model. We describe the data sets analyzed in this work in Section. 3. Finally, we present our results in Sect. 4 and summarise our findings in Sect. 5.

2. Inference approach

2.1. Basic theory and signal definition

Conceptually, the standard ΛCDM model of cosmology rests upon the cosmological principle, the notion that matter is distributed homogeneously and isotropically on large scales. On these scales, matter is described as a perfect fluid following a simple equation of state (EOS),

$$p_i = w_i \rho_i, \tag{1}$$

where ρ_i, p_i and w_i are the energy density, pressure, and EOS parameter, respectively. In ΛCDM, the EOS parameter w is assumed to be a constant for each fluid i contributing to the universe's total energy density ρ:

$$\rho = \sum_i \rho_i = \sum_i \rho_{0i} a^{-3(1+w_i)}. \tag{2}$$

Here, ρ_{0i} is the energy density of the i-th contribution today and a is the scale factor related to the redshift through $a = (1+z)^{-1}$. Contributions may include ordinary and dark matter, radiation, dark energy, and the universe's intrinsic curvature, denoted by ρ_m, ρ_r, ρ_Λ, and ρ_k, respectively.

One may then solve the Einstein field equations to determine the universe's evolution through its expansion rate, quantified by the Hubble parameter,

$$H(a) = \frac{\dot{a}}{a}. \tag{3}$$

The expansion rate is related to the universe's total energy density via the first Friedmann equation:

$$H(a)^2 = \frac{8\pi G}{3}\rho = \frac{8\pi G}{3}\Big(\rho_{0m}a^{-3} + \rho_{0r}a^{-4} + \rho_{0k}a^{-2} + \rho_{0\Lambda}\Big). \tag{4}$$

Here, the dependence on the scale factor a is characteristic for each type of fluid, arising from different values of the EOS parameter w (e.g., $w = 0$ for matter, $w = 1/3$ for radiation and $w = -1$ for a constant dark energy). At a given redshift, there is usually one dominant contribution to the universe's total energy budget, defining different cosmological eras.

In this work, we do not assume individual components contributing to the total energy density ρ as their evolution through cosmic time is not known a priori. Instead, we define a signal s as the energy density on a double-logarithmic scale

$$s(x) := \ln\Big(\frac{\rho(x)}{\rho_0}\Big), \tag{5}$$

where $x = -\ln(a) = \ln(1+z)$ and ρ_0 is an arbitrary reference energy density.

For ΛCDM, the signal field $s_{\Lambda\mathrm{CDM}}(x)$ takes on the form of a piecewise linear function in these coordinates (e.g. in the radiation-dominated era, $s_{\Lambda\mathrm{CDM}}(x) \propto -4\ln(a)$ and $s_{\Lambda\mathrm{CDM}}(x) \propto 3x$ for the matter-dominated era). Each era corresponds to a linear function with a slope proportional to its EOS parameter, smoothly connecting to the linear function representing the next era.

Here, we model $s(x)$ as a nonparametric, stationary process $s_{\mathrm{dev}}(x)$ around a mean linear trend $s_{\mathrm{lin}}(x)$ (see Sect. 2.4), by imposing homogeneous statistics on the deviations $s_{\mathrm{dev}}(x)$, i.e. imposing that the covariance S^{dev} at two points x and y only depends on their difference:

$$S^{\mathrm{dev}}_{xy} = C_s(x-y),$$

with C_s being the 2-point-correlation function. This is justified because the nonparametric component only needs to capture oscillations around the mean linear trend. In practice, thanks to the log-log definition of the signal (Eq. (5)), the amplitude of these oscillations varies by roughly one order of magnitude at most. Therefore, a stationary process is well suited, since the variance $S^{\mathrm{dev}}_{xx} = C_s(0)$ is constant for all x.

2.2. Instrument response

To constrain the expansion history of the universe, one measures the redshifts of objects at cosmological distances. The way redshift varies with distance encodes information about how the universe has expanded over time.

The luminosity distance provides the distances from observed fluxes emitted by distant objects. In a flat universe and assuming the FLRW metric, the luminosity distance is given by

$$d_L(z) = (1+z)\int_0^z \frac{c}{H(z')}\,\mathrm{d}z'. \tag{6}$$

[1] `NIFTy` (numerical information field theory) is available at https://ift.pages.mpcdf.de/nifty/user/index.html

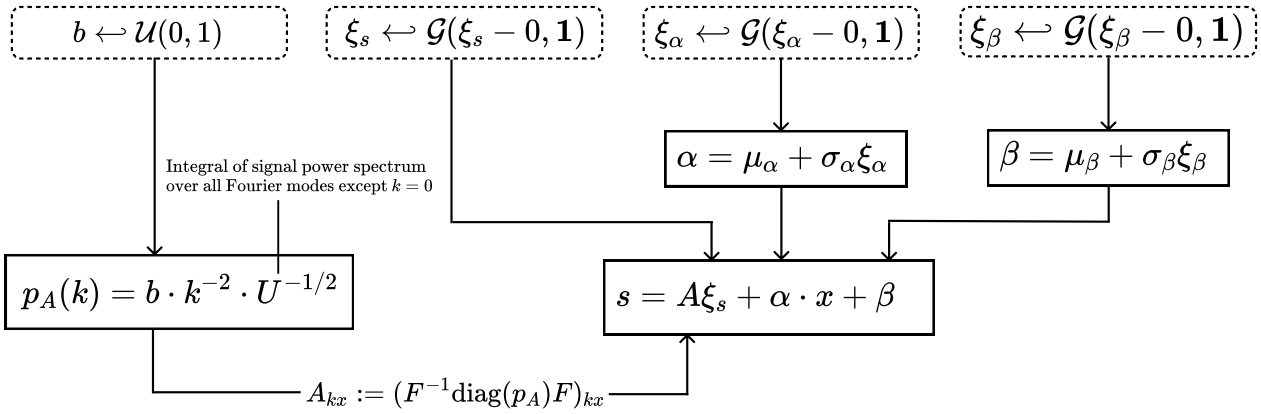


Fig. 1: The generative model used for the inference of the cosmic expansion history as encoded in $\ln(\rho/\rho_0)$ in charm2. The right side of the graph describes the generation of the linear part of the signal s in the charm2's model. The linearly evolving part of the signal, $\alpha x + \beta$, captures the dominant part of the density evolution and corresponds to the dominant cosmic constituent. The left-hand side of the graph describes the generation of the nonparametric $A\xi_s$ part of the signal, which accounts for deviations from a single-component equation of state. The random variates α, β, n and b are described by the distributions shown in Table 1.

Through a multi-step calibration process known as the cosmic distance ladder, the luminosity distance is related to the observed distance modulus of SNIA via

$$\mu = 5\log_{10}(d_L^{(\mathrm{pc})}) - 5, \tag{7}$$

where $d_L^{(\mathrm{pc})}$ denotes the luminosity distance in units of pc. These distance moduli constitute our discrete set of data d, which depend on the physical quantity $s = \log(\rho/\rho_0)$ we want to infer: the energy density determines the evolution of the Hubble parameter through the first Friedmann equation. The Hubble parameter in turn influences the luminosity distance via Eq. (6).

Through our signal definition Eq. 5, the Hubble parameter can be expressed as

$$H^2(x) = \frac{8\pi G}{3}\rho(x) = \frac{8\pi G}{3}\rho_0 e^{s(x)}. \tag{8}$$

We set the reference energy density ρ_0 to

$$\rho_0 = 10^6 \frac{\mathrm{kg}}{\mathrm{m} \cdot \mathrm{Mpc}^2}, \tag{9}$$

such that the Hubble-Parameter is in the usual units of km/s/Mpc: $[H(a)] = \sqrt{[G] \cdot [\rho]} = \sqrt{[G] \cdot \rho_0} = \mathrm{km/s/Mpc}$, where G is Newton's gravitational constant in SI-units.

In IFT, the function that maps the signal to the (noiseless) data is defined as the response operator. This function encapsulates the complex physical processes leading to the data generation. Using the SN Ia distance modulus, Eq. (7), together with Eq. (8) allows us to write the response operator as

$$R_x(s) = 5\log_{10}\left(e^x \mathcal{K} \int_0^x e^{-\frac{1}{2}s(\tilde{x})+\tilde{x}}\, \mathrm{d}\tilde{x}\right) - 5, \tag{10}$$

where $\mathcal{K}$ is a constant defined as

$$\mathcal{K} = 1000 \cdot c \left(\frac{8\pi G}{3}\right)^{-\frac{1}{2}}, \tag{11}$$

with c and G implicitly given in SI-units. The constant $\mathcal{K}$ is chosen such that the requirement of the luminosity distance, as expressed in Eq. (7), to be in pc, is met: the line-of-sight integral is implicitly an integral over $c/H(x)$ which is in $1000 \cdot \mathrm{pc}$ since $H(x)$ is in km/s/Mpc. Expressing $H(x)$ in terms of the signal field $s(x)$ through Eq. (8) introduces the additional $8\pi G/3$ factor. We will write $\hat{H}_0$ to indicate a Hubble constant implicitly given in km/s/Mpc, instead of writing out the unit explicitly.

Assuming additive noise, n, the data can be written as

$$d = R(s) + n =: s' + n, \tag{12}$$

where s' is referred to as the noiseless data.

2.3. Inference method

Our inference method recovers the posterior distribution of the signal given the data, $\mathcal{P}(s\,|\,d)$, which is given by Bayes' Theorem as

$$\mathcal{P}(s\,|\,d) = \frac{\mathcal{P}(d\,|\,s)\mathcal{P}(s)}{\mathcal{P}(d)}, \tag{13}$$

where $\mathcal{P}(d\,|\,s)$ is the likelihood, $\mathcal{P}(s)$ is the signal prior and the normalisation $\mathcal{P}(d)$ is the evidence. In IFT, the evidence is redefined as a partition function,

$$\mathcal{Z}(d) := \mathcal{P}(d) \tag{14}$$

and negative log-probabilities as information Hamiltonians,

$$\mathcal{H} = -\ln(\mathcal{P}). \tag{15}$$

Then, Bayes' Theorem is given by

$$\mathcal{P}(s\,|\,d) = \frac{1}{\mathcal{Z}} e^{-\mathcal{H}(s,d)} = \frac{1}{\mathcal{Z}} e^{-\mathcal{H}(d\,|\,s)-\mathcal{H}(s)}. \tag{16}$$

To define the likelihood $\mathcal{H}(d\,|\,s)$, we assume a Gaussian distribution based on the Maximum Entropy principle. Therefore, $\mathcal{H}(d\,|\,s)$ is given by:

$$\mathcal{H}(d\,|\,s) \mathrel{\widehat{=}} \frac{1}{2}(d - R(s))^\dagger N^{-1}(d - R(s)), \tag{17}$$

where $\widehat{=}$ denotes equality up to constants.

We assume that the matter distribution in the universe is homogeneous and isotropic on large scales and that the total energy density ρ evolves smoothly with the scale factor, while it

potentially may vary over multiple orders of magnitude. As discussed in Sect. 2.1, the signal definition $s(x) = \ln(\rho(x)/\rho_0)$ with $x = -\ln(a)$ ensures that the amplitude of these variations is comparable among the different eras of cosmic history. A homogeneous process with respect to x is then well-suited, since its covariance is given by

$$S^{\mathrm{dev}}_{xy} = C_s(x - y), \tag{18}$$

where C_s is again the signal's 2-point-correlation function. This leads to the variance being constant over the x-domain, $S^{\mathrm{dev}}_{xx} = C_s(0)$, reflecting that s exhibits fluctuations of comparable amplitude across all eras in the x-coordinate.

Thus, the prior term $\mathcal{H}(s)$ is defined as a quadratic form with a stationary covariance matrix as in Eq. (18). Rather than storing a dense matrix, we apply the Wiener-Khinchin theorem, stating that a covariance matrix becomes diagonal in Fourier space[2]:

$$\tilde{S}^{\mathrm{dev}}_{kq} = 2\pi \cdot \delta(k - q) p_s(k), \tag{19}$$

where the diagonal elements are given by the signal power spectrum $p_s(k)$.

Following the argumentation of Porqueres et al. (2017), the a priori signal power spectrum $p_s(k)$ is given by

$$p_s(k) = \nu k^{-4}, \tag{20}$$

where ν determines the amplitude of field fluctuations. The exponent of -4 corresponds to the total curvature of the field,

$$C := \int \mathrm{d}x\, |\Delta s(x)|^2, \tag{21}$$

which taking the scalar product to be the L^2-norm, it can be written as $C = s^\dagger(\Delta^\dagger\Delta)s$. Identifying $S^{-1} := \nu^{-1}\Delta^\dagger\Delta$ and Fourier-transforming leads to the exponent -4 in Eq. (20).

Thus, S^{-1} takes on the function of a smoothing kernel, and the constant ν is a scaling constant that determines how much the probability of highly curved signal realizations is weighed down in the inference. This constant was set to $\nu = 1$ in Eq. (20) in Porqueres et al. (2017). Here, while we assume the same power-law dependence, we directly infer the smoothing constant ν from the data. More specifically, we replace the fixed parameter ν with a related quantity, the fluctuation parameter b, which captures fluctuations around a mean trend (see Sect. 2.4). In contrast to Porqueres et al. (2017), where the field variance at each point scales with ν as $\langle s^x s^{x^*}\rangle_{(s)} = \nu \int_{k>0} k^{-4}\mathrm{d}k$, our model directly sets the variance as $\langle s^x s^{x^*}\rangle_{(s)} = b^2$.

Given the data, prior and likelihood terms, we only need to apply Bayes' Theorem (Eq. (16)) to obtain the posterior distribution of the signal. However, analytical solutions are often not possible and, in many cases, the partition function $\mathcal{Z}$ is intractable analytically or computationally prohibitive (Knollmüller & Enßlin 2018).

Variational inference (VI) addresses this problem. The posterior distribution $\mathcal{P}(s\,|\,d)$ is approximated through a trial distribution, using the Kullback-Leibler (KL) divergence as a measure of dissimilarity between probability distributions.

Knollmüller & Enßlin (2020) use a Gaussian for the trial distribution. However, the posterior is often non-Gaussian, leading to a sub-optimal approximation. In this work, we employ the `geoVI` algorithm (Frank et al. 2021), which finds a coordinate transformation that effectively absorbs potential curvature of the posterior and finds a new coordinate system where the posterior is well approximated by a Gaussian distribution (see details in App. A).

2.4. Generative model prior

As prior information, we assume that the universe evolves monotonically (expanding, contracting, or static) in a smooth fashion. The monotonicity of the evolution is captured through a linear model:

$$s_{\mathrm{lin}} = \alpha x + \beta, \tag{22}$$

where α and β are random variables drawn from a Gaussian prior. Deviations from this linear evolution are modelled via a Correlated Field Model (CFM). The CFM, introduced in Arras et al. (2021) and Arras et al. (2022), generates field realizations from a parametric power spectrum with contributions from stochastic processes:

$$s_{\mathrm{CFM}} = A\xi_s, \tag{23}$$

where A is the so-called amplitude operator and ξ_s is a Gaussian standard variable, $\xi_s \hookleftarrow \mathcal{G}(0, \mathbf{1})$. In Fourier space, the diagonal elements of the amplitude operator are given by the amplitude spectrum $p_A(k)$, which is related to the signal power spectrum via $p_A(k) = \sqrt{p_s(k)}$ (see App. A).

In this work, we will assume that the amplitude spectrum follows a power-law:

$$p_A(k) = b \cdot \frac{k^n}{\sqrt{U}}, \tag{24}$$

where $U = \int_{k\neq 0} \mathrm{d}k\, k^{2n}$ normalizes the amplitude spectrum such that the real-space variance of the field is given by the fluctuations parameter b: $\langle s^x s^{x^*}\rangle = \int_{k\neq 0} \mathrm{d}k\, p_A(k)^2 = b^2$.

Adding the contribution by the linear model and the CFM gives us a signal realization:

$$s = s_{\mathrm{lin}} + s_{\mathrm{CFM}} = \alpha x + \beta + A\xi_s. \tag{25}$$

A flowchart of this forward model can be seen in Fig. 1.

Table 1 shows the names, distributions, and prior mean and standard deviation of the hyperparameters. Here we discuss these prior choices:

- α: the prior mean slope of the line model is set to 2. The standard deviation is set to the same order of magnitude to ensure sufficient model flexibility.
- β: the prior mean is chosen as the $s(x = 0)$ value of flat ΛCDM cosmology fields (cf. Fig. 6), again with a standard deviation in the same order of magnitude
- n: is set to -2 according to Eq. (20). Notice that this is a factor of $1/2$ smaller than the exponent found in Eq. (20) to account for the fact that $p_s(k) = \sqrt{p_A(k)}$
- b: drawn from a uniform distribution $\mathcal{U}$ in the interval $[0, 1]$ to be maximally agnostic (see Fig. 2). The order of magnitude of this parameter is obtained by comparing to a reference value, $b_{\Lambda\mathrm{CDM}}$, the expected fluctuations in the case of a flat ΛCDM cosmology. This reference value was obtained through a linear best-fit to a reference field constructed using the Planck2018 (Planck Collaboration 2020) cosmology parameters and found to be $b_{\Lambda\mathrm{CDM}} \approx 0.14$. Therefore, drawing b from $\mathcal{U}(0, 1)$ should allow for enough flexibility to reconstruct any non-ΛCDM features if preferred by the data.

[2] In our notation, we use $\tilde{f}(k) = \int \mathrm{d}x\, e^{ikx} f(x)$ for the forward and $f(x) = \int \mathrm{d}k/(2\pi)\, e^{-ikx}\tilde{f}(k)$ for the backward Fourier transform, where k is the wavevector associated with the real space variable $x = -\ln(a)$.

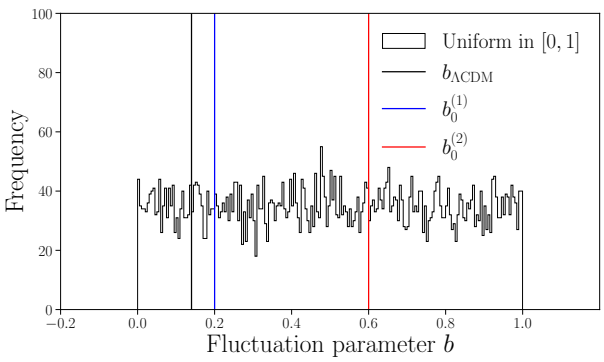


Fig. 2: Prior samples for the fluctuations parameter, b, which are drawn from a uniform distribution in the interval $[0, 1]$. This parameter sets the a priori expected level of field fluctuations around the field mean. We use the fluctuations parameter expected from a flat ΛCDM model, $b_{\Lambda CDM}$ (black vertical line), to set an appropriate scale for this distribution heuristically (see App. B). The blue and red vertical lines indicate two possible starting positions for the inference scheme, which were used in the real data reconstructions of this work.

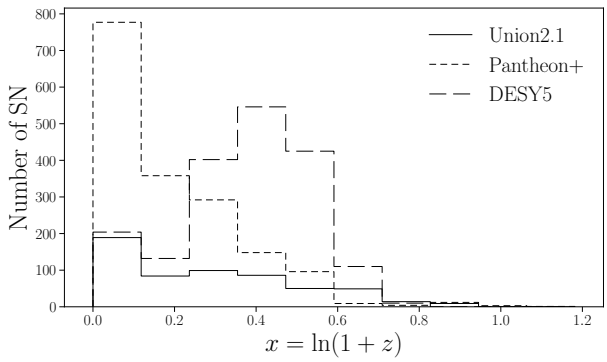


Fig. 3: Histograms of the datasets binned in ten x bins. While the Pantheon+ sample dominates the low redshift regime, the new DESY5 sample shows a significant boost in the number of mid-to-high redshift SN samples.

The CFM enforces periodic boundary conditions via Fast Fourier transforms of the power spectrum. To avoid artefacts due to periodic boundary conditions, we double the signal domain and introduce a zero-padding mask in the margins (Arras et al. 2021).

Table 1: Hyperparameters chosen for the generative prior model. Gaussian and uniform distributions are denoted by $\mathcal{G}(\mu', \sigma')$ and $\mathcal{U}(l', u')$, where μ', σ', l' and u' are the mean, standard deviation, lower and upper bound respectively.

Description	Symbol	Distribution
Slope of line model	α	$\mathcal{G}(2, 5)$
Offset of line model	β	$\mathcal{G}(30, 10)$
Power law exponent of amplitude spectrum	n	$\mathcal{G}(-2, 10^{-16})$
Fluctuations	b	$\mathcal{U}(0, 1)$

3. Data

We use our inference method to analyze three main SN Ia data sets: Union2.1 (Suzuki et al. 2012), Pantheon+ (Scolnic et al. 2022) and DESY5 (Dark Energy Survey Year 5, Vincenzi et al. 2024; DES Collaboration 2024). We further discuss the impact of a recent recalibration of DESY5, dubbed "DES-Dovekie" Popovic et al. (2026), on our reconstruction results.

The data sets consist of distance moduli μ obtained by fitting the SN light curves with spectral energy distribution templates (Kenworthy et al. 2021),

$$\mu = m + \chi x_1 - \psi y - M - \delta_{\mu-\text{bias}}, \tag{26}$$

where $m = -2.5\log_{10}(x_0)$ is the apparent magnitude and M is the absolute magnitude; x_0, x_1 and y parametrize the lightcurve; and χ, ψ and $\delta_{\mu-\text{bias}}$ are calibration parameters.

For the distance modulus, the absolute magnitude M is degenerate with H_0. Therefore, many analyses combine the two parameters as (e.g., Perlmutter et al. (1999), Rubin et al. (2023) and DES Collaboration (2024)),

$$\mathcal{M} = M + 5\log_{10}(c/H_0), \tag{27}$$

which is jointly inferred with the cosmological parameters. In this case, the distance moduli can be obtained only by assuming a value for H_0.

Union2.1 and DESY5 assume a value of $h := H_0/(100\,\text{km/s/Mpc}) = 0.7$, while Pantheon+ uses Cepheids as a primary distance anchor to infer a value of H_0 from low-redshift measurements, allowing to lift the degeneracy in Eq. 27.

Figure 3 shows the redshift distributions of the main data sets, and Table 2 indicates their redshift range.

Table 2: Redshift range and number of datapoints of the data sets analyzed in this work.

Data set	$(\mathbf{z}_{\min}, \mathbf{z}_{\max}, \bar{\mathbf{z}})$	# Datapoints
Union2.1	$(0.015, 1.414, 0.36)$	580
Pantheon+	$(0.00122, 2.26, 0.22)$	1701
DESY5	$(0.025, 1.12, 0.46)$	1829

4. Results

In this Section, we apply the charm2 algorithm to synthetic and real data, to validate our inference method and reconstruct the posterior mean field $s = \ln(\rho/\rho_0)$ from observations. Using the evidence lower bound ("ELBO") measure, often encountered in variational approaches, we determine whether there exists a preference for non-ΛCDM features in any data sets analyzed here. Finally, we discuss the effect of the recalibrated DESY5-Dovekie sample on our reconstructions.

4.1. Impact of b initialization on field reconstruction

To improve the convergence of the algorithm, a reasonable initial parameter vector must be chosen. We find that, in practice, the reconstructed posterior mean field depends on the choice of the initial fluctuation parameter, b_0, although reconstructions with differing initial b_0 values generally agree with each other within their reconstruction uncertainty.

Since the data values are line of sight integrals over $x = -\ln(a)$, signal fields that oscillate at a high rate around their mean line give approximately the same data as signals with damped oscillations. Therefore, SNe alone are a suboptimal probe to constrain the field fluctuations of $s = \ln(\rho/\rho_0)$.

We thus present our results in terms of the initial fluctuation parameter b_0, showing reconstructions where it was initialized at lower values, $b_0^{(1)} = 0.2$ and higher values, $b_0^{(2)} = 0.6$ (see vertical colored lines in Fig. 2). A first estimate of the field fluctuations can be obtained by fitting a linear model to the benchmark ΛCDM curves seen in Fig. 6. This provides a value for the residual variance, $b_{\Lambda\mathrm{CDM}} = 0.14$, on which we base our suggestion for the initial values: $b_0^{(1)} = 0.2$ lies close to the theoretical $b_{\Lambda\mathrm{CDM}}$ value while providing some additional flexibility, whereas $b_0^{(2)} = 0.6$ represents a significant potential deviation from ΛCDM.

4.2. Synthetic reconstructions

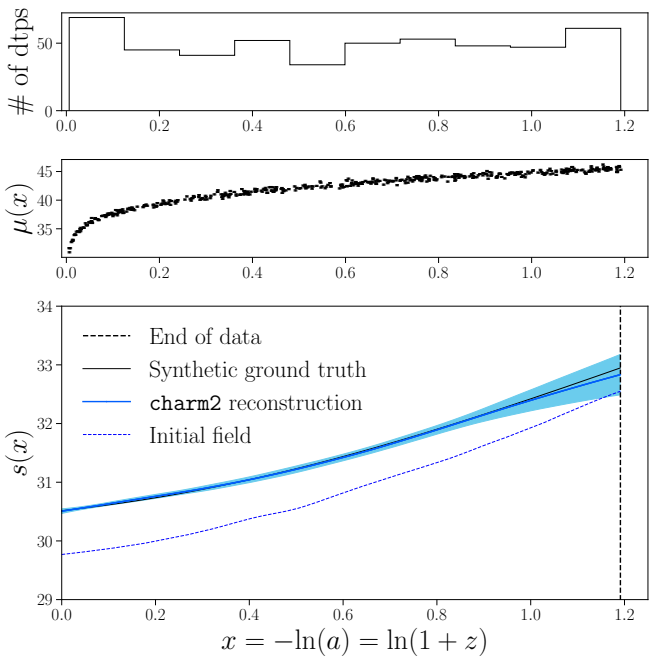


Fig. 4: Reconstruction from a synthetic data set with charm2. From top to bottom: Histogram of the number of data points; the data plotted as a function of the x-coordinate; and the initial field (dashed blue line) along with the posterior mean field (solid blue line) with its 1σ uncertainty band (shaded blue region), and the ground truth (solid black line). The inference was initialized with a fluctuation parameter of $b_0 = 0.2$. The noise variance in the simulated data is set to $\sigma_n^2 = 0.1$, with the errorbars vanishingly small compared to the scale of the y-axis.

We validate the inference method with synthetic data sets generated through a flat ΛCDM model with $\Omega_{\mathrm{m}0} = 0.3$. Each data set contains 500 datapoints, but the distribution over redshift varies as:

1. Data set 1: the redshifts are sampled from a uniform distribution within the redshift range, $\mathcal{P}(z) = \mathrm{cst}$.
2. Data set 2: the density of data points decays with redshift exponentially. The redshifts are sampled from the normalized distribution $\mathcal{P}(z) = \frac{1}{2}e^{-\frac{1}{2}z}$ and rescaled to the corresponding redshift range.

The redshifts and signal fields are then mapped into the data space through the response operator, Eq. (10), and Gaussian noise is added to mimic the dispersion of the data points.

The noise variance was set to 0.1, motivated by the mean variance of data points in the Union2.1 compilation (≈ 0.07).

For an initial fluctuation parameter of $b_0 = 0.2$, as shown in Fig. 4 (data set 1) and 5a (data set 2), our inference method recovers the true signal across the redshift range for both synthetic data sets. For data set 2 (Fig. 5a), the reconstruction uncertainty increases at high x, where fewer data points are available.

By comparing Fig. 5a ($b_0 = 0.2$) and 5b ($b_0 = 0.6$), we can assess that the analysis of synthetic data is robust against different initializations of the fluctuation parameter as the posterior mean fields agree within their reconstruction uncertainty.

4.3. Union2.1, Pantheon+ and DESY5 reconstructions

We applied charm2 to the Union2.1, Pantheon+ and DESY5 data sets. We compare our reconstructions to flat ΛCDM reference fields shown in Fig. 6, s_{CMB} and s_{SN}, corresponding to the signal for the cosmological parameters found by the Planck2018 (Planck Collaboration 2020) and the Pantheon+SH0ES (Brout et al. 2022) analyses, respectively (see App. B for construction).

The reconstructed fields for the three data sets are presented in Fig. 7, alongside histograms of their respective redshift distributions.

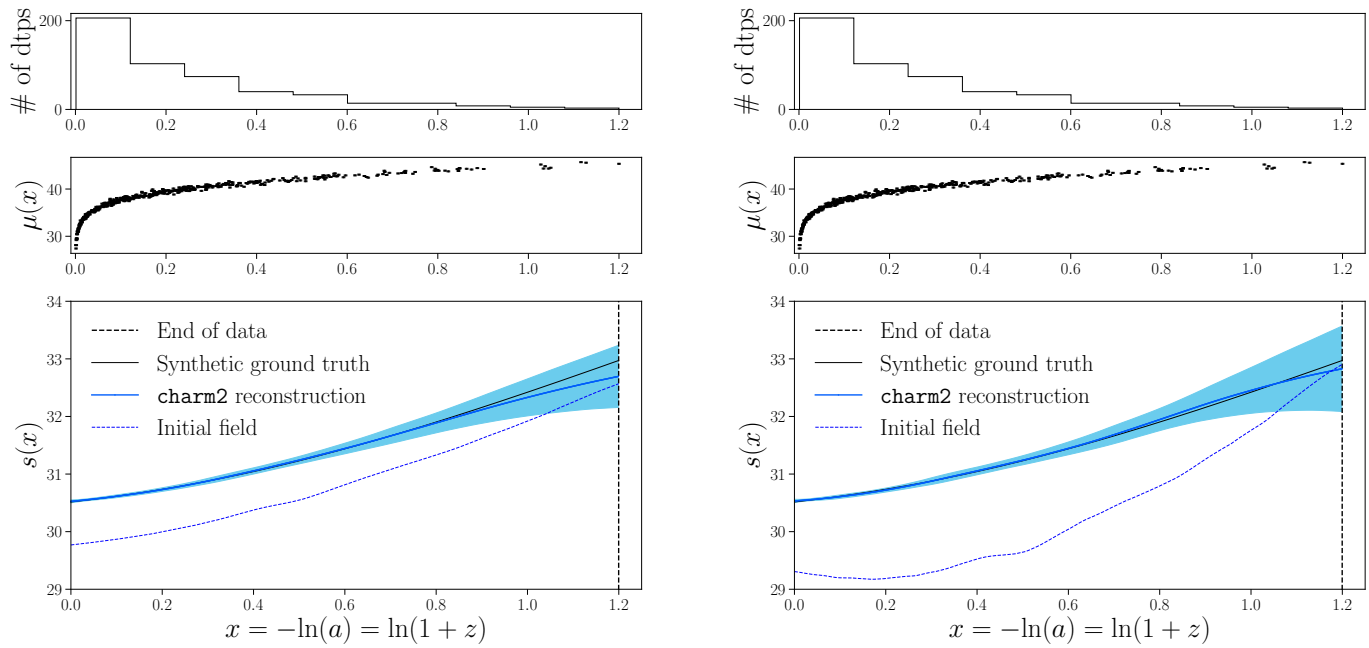
Union2.1

In the first data set to be analyzed, we compare charm2 to its predecessor code, charm1, which was only applied to Union2.1 data, as this was the most up-to-date SNIa sample at that time. While the reconstructed fields are compatible, charm2 provides more realistic uncertainties by better accounting for the nonlinearities of the response operator, which were neglected in Porqueres et al. (2017). As a result, we see that the reconstruction uncertainty at high redshifts was significantly underestimated in the charm1 reconstruction, demonstrating the advantage of non-Gaussian posterior approximation techniques. We also show the effect of changing charm2's initial fluctuations parameter from $b_0 = 0.2$ to $b_0 = 0.6$. The results are almost identical, indicating that the charm2 analysis of the Union2.1 data set is robust to the choice of the proposed b_0 initial values.

Note that the Union2.1 data assumes a fiducial value of $H_0 = 70$ km/s/Mpc to calibrate the distance modulus μ (see Sect. 3). This H_0 value affects the reconstruction at $z = 0$, making it lie closer to the s_{CMB} field. Since s_{CMB} however assumes $H_0 = 67.37$ km/s/Mpc, the two curves are slightly offset from each other.

Pantheon+

The charm2 analysis of Pantheon+ data is also robust to the choice of initial b_0 value, providing reconstructed fields that are compatible within the 1σ uncertainty, with slightly larger fluctuations around the mean line in the $b_0 = 0.6$ reconstruction. Further, the Pantheon+ reconstruction is compatible with the s_{SN} field, but deviates from s_{CMB} by more than 2σ at $x < 0.6$, a manifestation of the Hubble Tension, as discussed, e.g., in Riess et al. (2022).



(a) Initial fluctuations parameter $b_0 = 0.2$.

(b) Initial fluctuations parameter $b_0 = 0.6$.

Fig. 5: Same as Fig. 4 with an exponentially falling redshift distribution and with an initial fluctuation parameter of $b_0 = 0.2$ (left) and $b_0 = 0.6$ (right).

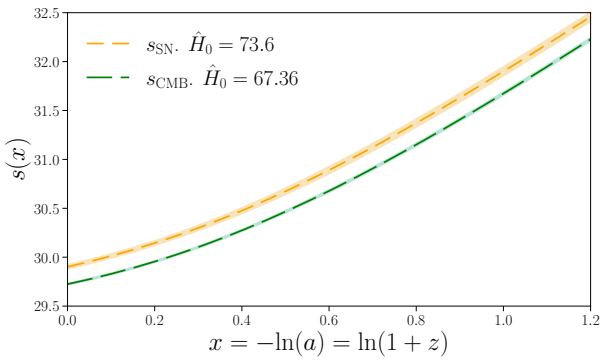


Fig. 6: Mean field and estimated 1σ region for the flat ΛCDM models with the Ω_m and H_0 values from Planck2018 (green) and Pantheon+SH0ES (orange). The 1σ contour was estimated through Gaussian error propagation of the uncertainties of the values in Table B.1 into Eq. (B.2), which assumes H_0 and Ω_m are independent. These fields serve as benchmarks for our reconstructions.

DESY5

The final data set analyzed in this work is the DESY5 SNIa sample, which has been reported to slightly prefer an evolving dark energy (EDE) component that follows an equation of state

$$w_\Lambda(a) = w_0 + w_a(1-a) \tag{28}$$

where w_0 sets the baseline and w_a controls the time evolution (DES Collaboration 2024). Using this model, often referred to as w_0w_aCDM, DES Collaboration (2024) found

$$(\Omega_m, w_0, w_a) \approx (0.495, -0.36, -8.8) \tag{29}$$

for the mean best-fit parameters. Using a fiducial Hubble constant of $H_0 = 68$ km/s/Mpc[3], we calculate the parametric signal field expected from this EDE scenario, $s_{\rm EDE}$, and use it as an additional benchmark for the DESY5 reconstructions.

The charm2 reconstruction of the original DESY5 data shows a feature at $x \sim [0.45, 0.75]$ (corresponding to redshift $z \sim [0.57, 1.17]$) that is in tension with both the $s_{\rm SN}$ and the $s_{\rm CMB}$ flat ΛCDM fields, while in agreement with the evolving dark energy $s_{\rm EDE}$ field for both the $b_0 = 0.2$ and $b_0 = 0.6$ initial conditions, as well as a very low initial fluctuation value of $b_0 = 0.05$. The influence of the initial b_0 value on the posterior mean field is most clearly seen in the analysis of the DESY5 data. Note however that a value of $b_0 = 0.05$ leads to posterior samples that only barely fluctuate around their mean linear trend, which in effect corresponds to a linear fit of the expansion history and is therefore suboptimal from a theoretical point of view. Therefore, higher values such as $b_0 = 0.2$ and $b_0 = 0.6$, should be considered more reliable.

We tested whether an evolving-dark-energy-like feature could be caused by a systematic offset in magnitude between low- and high-redshift SNe in the sample, even though the ground truth is flat ΛCDM. This is motivated by the systematic offset of 0.04 mag pointed out by Efstathiou (2025) between the low-and high-redshift data of the Pantheon+ and DESY5 compilations. For the necessary mock study, we generated data according to a flat ΛCDM model and introduced a shift of +0.1 mag for all data points falling under an arbitrary threshold $x < 0.6$. Figure 8 shows that in such a case, the reconstructed mean field exhibits a deviation visually similar to the one observed in our DESY5 reconstructions at high redshifts, indicating that systematics might cause the observed feature. However, we could

[3] Through a variational inference w_0w_aCDM fit to the DESY5 data, we found a posterior value of $\hat{H}_0 = 68.1 \pm 0.4$ and thus heuristically set $\hat{H}_0 \approx 68$

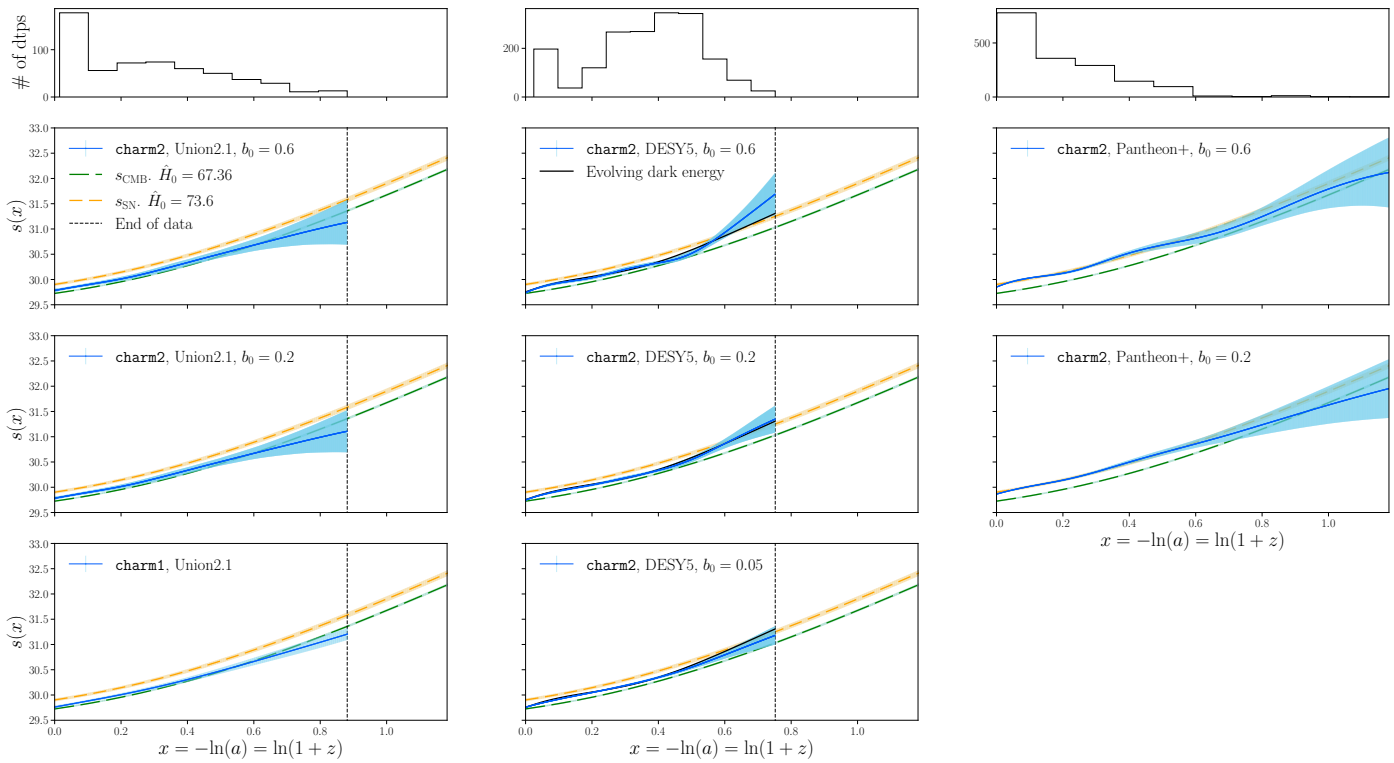


Fig. 7: charm2 reconstructions for Union2.1 (left column), DESY5 (middle column) and Pantheon+ (right column) and for different initial values of the fluctuation parameter b_0 (rows). In each panel, the posterior mean-field (solid blue line) and its 1σ confidence region (shaded blue region) are displayed alongside the reference fields of Fig. 6 (dashed orange and green lines). Additionally, for the DESY5 reconstruction, the signal field corresponding to the evolving dark energy model suggested by DES Collaboration (2024) ($(\Omega_m, w_0, w_a) = (0.495, -0.36, -8.8)$) is shown as the black solid line. A comparison with the previous charm1 analysis of Union2.1 is shown in the left column (lower panel), where the signal definition is adapted as described in App. C.

not identify a systematic offset at low redshift $x < 0.1$ that reproduces such a feature, which is the regime where Efstathiou (2025) found $\langle m_{\mathrm{Pantheon+}} - m_{\mathrm{DESY5}} \rangle = (-0.051 \pm 0.007)$ mag. Furthermore, Vincenzi et al. (2025) clarified that the discrepancy between DESY5 and Pantheon+ arises from changes in the DESY5 analysis pipeline, leaving a potential systematic offset of the order of 0.008 mag. Our tests showed that such a systematic offset would no longer have a significant impact on our reconstruction results, in which case the reconstructions would be compatible with an evolving dark energy scenario.

Further, Popovic et al. (2026) recently introduce an updated version of DESY5, making improvements on the photometric calibration, error handling and the treatment of dust reddening. We reran the charm2 analysis on this updated data set and find that for $b_0 = 0.05$ and $b_0 = 0.2$ the DESY5-Dovekie reconstructions change only marginally compared to their DESY5 counterparts in Fig. 7. In the more flexible $b_0 = 0.6$ case, we find that the energy values of the posterior mean field are significantly reduced at high redshifts when going from DESY5 to DESY5-Dovekie, cf. Fig. 9. This deexcitation of the posterior mean field of the nonparametric reconstruction can therefore be attributed to the corrected systematics in the DESY5-Dovekie catalog. However, it remains to be assessed whether any nonparametric excitations of the signal field are genuinely preferred by the data.

4.4. Model selection

The evidence term in Bayes' Theorem,

$$\mathcal{P}(d) = \int \mathrm{D}s \, \mathcal{P}(d|s)\mathcal{P}(s), \tag{30}$$

where $\int \mathrm{D}s$ denotes an integral over all s field configurations, gives the probability that the observed data was generated by the assumed prior model. It is thus a useful tool for selecting between two physical models that are consistent with the data, with the model producing the higher evidence term being preferred. However, in our high dimensional, nonparametric setting, the integration in Eq. (30) is computationally expensive and we instead use a proxy for the evidence known as the evidence lower bound. The ELBO can be calculated by averaging the likelihood $\mathcal{H}(d|s)$ over the approximate posterior samples q found by geoVI and taking the difference with an analytic term A:

$$\mathrm{ELBO} = -\langle \mathcal{H}(d|s) \rangle_q + A. \tag{31}$$

The analytic term is included in App. D. There, we also note that this is a lower bound on the log evidence

$$\ln(\mathcal{P}(d)) \geq \mathrm{ELBO}. \tag{32}$$

These two quantities differ by the information difference in nits between the posterior and its variational approximation (Guardiani et al. 2022). If the variational approximation captures the true posterior well, the ELBO is approximately equal to the log evidence. Given the complexity of distributions geoVI can

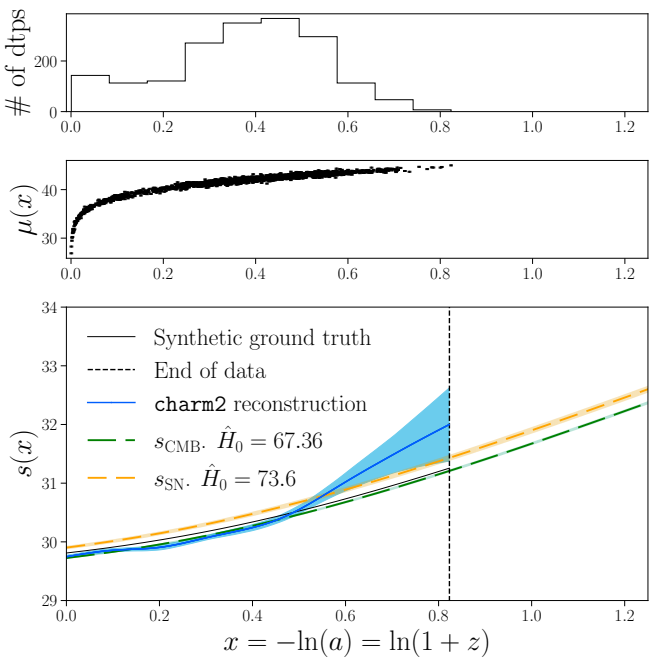


Fig. 8: A $b_0 = 0.6$ charm2 reconstruction for a synthetic data set generated with a flat ΛCDM model with $\Omega_{m0} = 0.3$ and $H_0 = 70.3$ km/s/Mpc, where all data points $x < 0.6$ were systematically shifted by +0.1 mag.

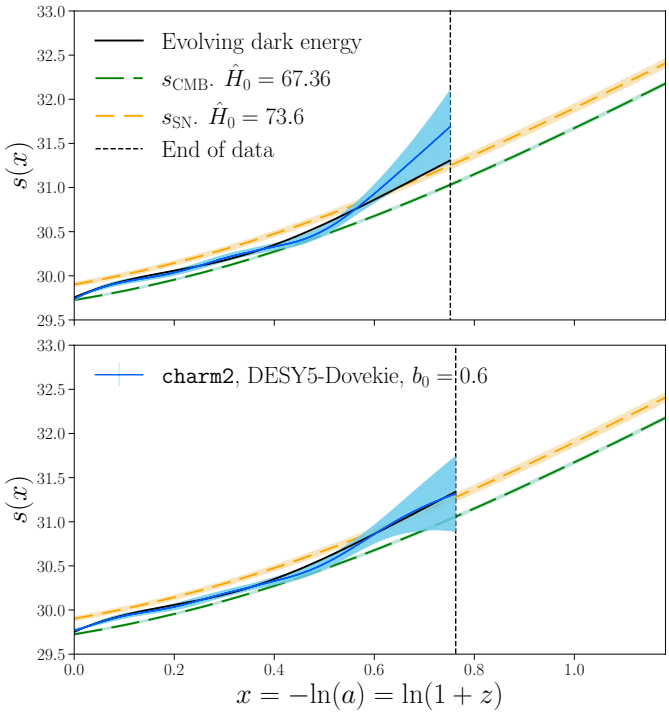


Fig. 9: A comparison between the $b_0 = 0.6$ charm2 reconstructions for the original DESY5 and the updated DESY5-Dovekie data set.

represent, we find the ELBO to be a very good proxy for the evidence.

Note that Eq. (31) naturally introduces a sampling uncertainty which we quote in Tab. 3, where the ELBOs computed from flat ΛCDM, flat w_0w_aCDM and our nonparametric model fit are displayed for all data sets. Using the ELBO, we find no conclusive evidence for non-ΛCDM features in the SNIa samples analyzed in this work.

This is likely due to flat ΛCDM being able to explain the data extremely well with only 2 parameters (H_0 and Ω_m), while the extra parameters added in flat w_0w_aCDM and our nonparametric model do not increase goodness of fit significantly enough to counterbalance the penalty by the Occam's Razor term (see Eq. D.6).

Note, however, that the ELBO might be a suboptimal model selection statistic due to an insufficient signal-to-noise ratio. In a synthetic study using the DESY5 noise covariance matrix, we compared the flat ΛCDM model and a $b_0 = 0.6$ charm2 reconstruction with an evolving dark energy scenario as ground truth, see Fig. 10. While the nonparametric reconstruction encloses 68% of the ground truth within its 1σ uncertainty bands, the flat ΛCDM fit encloses only 13%, leading to a confident rejection of the evolving dark energy ground truth. The evidences in this scenario,

$$\mathrm{ELBO}_{\text{flat }\Lambda\mathrm{CDM}} = -894.19 \pm 0.85 \quad (33)$$

$$\mathrm{ELBO}_{\text{non-param.}} = -905.5 \pm 1.8 \quad (34)$$

support this overconfident rejection of evolving dark energy. While including other cosmology probes might increase the reliability of the ELBO, at the current noise level of the DESY5 survey, we find that for SNIa alone the noise covariance would need to be smaller by a factor of ~ 0.1 for the ELBO diagnostic to start preferring the nonparametric model which more accurately represents the ground truth over standard flat ΛCDM (see Fig. 11). If the ELBO is a sufficiently accurate proxy, this argumentation extends to the evidence itself and thus underscores the importance of synthetic control studies to judge its reliability when it comes to model selection.

To conclude, we find no data-driven evidence favoring nonparametric deviations from flat ΛCDM in the data sets analyzed in this work, which are the Union2.1, Pantheon+, DESY5 and DESY5-Dovekie supernova samples. At the same time, we caution that for SNIa alone, the evidence might be a sub-optimal model selection statistic for nonparametric models at current noise levels.

5. Conclusions

We have presented charm2, a Bayesian method to reconstruct the cosmic expansion from supernova data using information field theory. charm2 improves over charm1 (Porqueres et al. 2017) by providing more accurate uncertainties and a better treatment of the nonlinearities in the data, both of which are achieved by using the geometric variational inference algorithm (Frank et al. 2021). charm2 also uses a flexible Bayesian hierarchical model that permits inference of central quantities, such as the mean background cosmology and the statistical power spectrum, jointly with the quantities of interest. This ensures that charm2 is robust against biases arising from assuming a cosmological model.

We tested the effect of differing fluctuation parameter initializations b_0, which regulate the expected deviations from the mean background cosmology. We demonstrated that the analyses of Union2.1 and Pantheon+ data were robust against the initial choice of b_0 and the inferred reconstructions are compatible with flat ΛCDM.

Table 3: ELBO of all charm2 reconstructions shown in Fig. 7 compared against flat ΛCDM and flat w_0w_aCDM fits of the data. All errors were rounded to two significant figures.

	Union2.1	**DESY5**	**DESY5-Dovekie**	**Pantheon+**
Flat ΛCDM	−278.9 ± 1.2	−829.02 ± 0.77	−824.8 ± 1.3	−885.2 ± 1.2
Flat w_0w_aCDM	−282.2 ± 6.5	−829.0 ± 1.2	−826.0 ± 1.4	−886.0 ± 1.3
nonparametric charm2				
$b_0 = 0.6$	−284.3 ± 1.3	−835.0 ± 2.6	−832.1 ± 3.9	−889.3 ± 3.1
$b_0 = 0.2$	−284.2 ± 1.2	−834.1 ± 2.3	−830.2 ± 2.2	−889.6 ± 3.6
$b_0 = 0.05$		−834.6 ± 2.5	−831.0 ± 2.3	

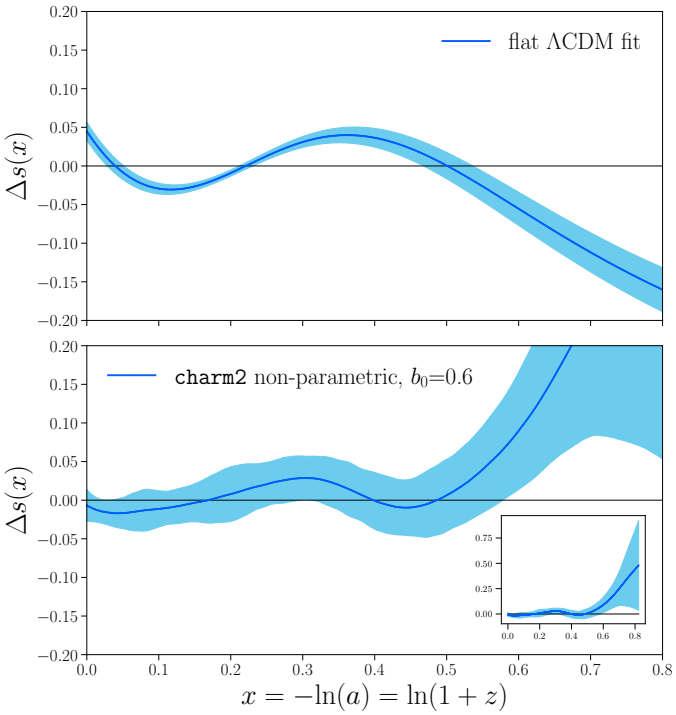


Fig. 10: Residuals between an evolving dark energy ground truth and a flat ΛCDM fit (upper panel) and a nonparametric $b_0 = 0.6$ charm reconstruction (lower panel). The inset axis shows a zoomed out version of the nonparametric reconstruction. The synthetic study used the real DESY5 survey noise covariance for the generation of synthetic data.

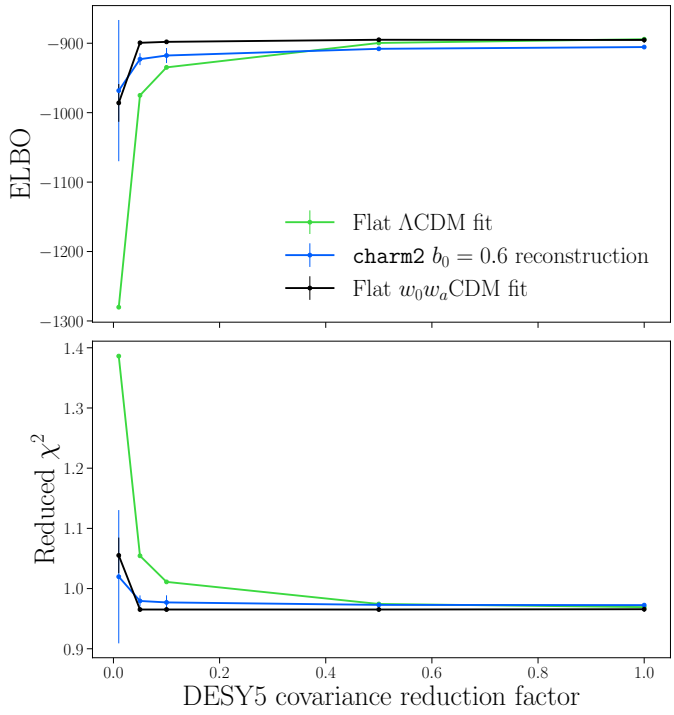


Fig. 11: Evolution of the ELBO and reduced likelihood χ^2 with varying noise level. The synthetic data was generated according to an evolving dark energy ground truth and the real DESY5 survey noise covariance matrix. For each point, the full noise covariance was multiplied by a constant factor, given by the corresponding x-axis value (high signal-to-noise on the left, low on the right).

The Union2.1 catalog, which assumes $H_0 = 70$ km/s/Mpc for data calibration, results in a reconstruction mostly compatible with Planck2018 (Planck Collaboration 2020), whereas the Pantheon+ reconstruction deviates significantly from Planck2018 at lower redshifts, a manifestation of the Hubble tension as discussed in Riess et al. (2022).

The reconstruction from DESY5 data showed a significant deviation from the flat ΛCDM model at $x \gtrsim 0.45$ ($z \gtrsim 0.57$) for initial fluctuation values of $b_0 \gtrsim 0.1$. Through a numerical experiment using synthetic reconstructions, we ruled out the possibility that this deviation could be explained by a systematic offset in the distance moduli in the redshift range suggested by Efstathiou (2025). We showed that our DESY5 reconstruction is compatible with an evolving dark energy signal. Going from the DESY5 catalog to its updated version, DESY5-Dovekie and choosing $b_0 = 0.6$, which potentially allows for larger deviations from flat ΛCDM, the resulting posterior mean field shifts toward lower energy at high redshift, effectively relaxing the field. This suggests that the nonparametric model was partially capturing systematics present in DESY5 that are mitigated in DESY5-Dovekie.

Using the evidence lower bound as a proxy for the evidence, we find that flat ΛCDM is consistently preferred over our nonparametric reconstructions. Accordingly, the supernova samples analyzed here do not provide evidence for non-flat-ΛCDM features that would be captured by a nonparametric model. However, we note that at current noise levels and with SNIa alone, the evidence lower bound and by extension the evidence might be a suboptimal model selection statistic.

In summary, we have developed a nonparametric, model-agnostic method to infer the cosmic history from SN data. This method improves the set of tools we can use to rationally question and investigate the impact of assumptions on data analysis.

Possible extensions of the charm2 method are the inclusion of other data sets, such as Cepheids, and the use of the SNe spatial distribution to constrain anisotropies of the expansion rate.

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References

Adame, A., Aguilar, J., Ahlen, S., et al. 2025, Journal of Cosmology and Astroparticle Physics, 2025, 021
Arras, Bester, Perley, et al. 2021, A&A, 646, A84
Arras, P., Frank, P., Haim, P., et al. 2022, Nature Astronomy, 6, 259–269
Benetti, M. & Capozziello, S. 2019, Journal of Cosmology and Astroparticle Physics, 2019, 008
Brout, D., Scolnic, D., Popovic, B., et al. 2022, The Astrophysical Journal, 938, 110
DES Collaboration. 2024, ApJ, 973, L14
Efstathiou, G. 2025, Monthly Notices of the Royal Astronomical Society, 538, 875
Enßlin, T. A., Frommert, M., & Kitaura, F. S. 2009, Physical Review D, 80
Frank, P., Leike, R., & Enßlin, T. A. 2021, Entropy, 23, 853
Freese, K. 2009, EAS Publications Series, 36, 113–126
Guardiani, M., Frank, P., Kostić, A., et al. 2022, PLOS ONE, 17, 1
Holsclaw, T., Alam, U., Sansó, B., et al. 2010, Phys. Rev. Lett., 105, 241302
Holsclaw, T., Alam, U., Sansó, B., et al. 2011, Phys. Rev. D, 84, 083501
Ishida, E. & de Souza, R. 2010, Astronomy and Astrophysics
Jiang, J.-Q., Pedrotti, D., da Costa, S. S., & Vagnozzi, S. 2024, Non-parametric late-time expansion history reconstruction and implications for the Hubble tension in light of DESI
Joy, M., Knox, L., & Aylor, K. 2019, in American Astronomical Society Meeting Abstracts, Vol. 233, American Astronomical Society Meeting Abstracts #233, 356.02
Kenworthy, W. D., Jones, D. O., Dai, M., et al. 2021, The Astrophysical Journal, 923, 265
Knollmüller, J. & Enßlin, T. A. 2018, Encoding prior knowledge in the structure of the likelihood
Knollmüller, J. & Enßlin, T. A. 2020, Metric Gaussian Variational Inference
Matthewson, W. L. & Shafieloo, A. 2024, arXiv e-prints, arXiv:2409.02550
Mukherjee, P., Said, J. L., & Mifsud, J. 2022, Journal of Cosmology and Astroparticle Physics, 2022, 029
Mukherjee, P. & Sen, A. A. 2024, Model-independent cosmological inference post DESI DR1 BAO measurements
Perlmutter, S., Aldering, G., Goldhaber, G., et al. 1999, ApJ, 517, 565
Planck Collaboration. 2020, A&A, 641, A6
Popovic, B., Shah, P., Kenworthy, W. D., et al. 2026, The Dark Energy Survey Supernova Program: A Reanalysis Of Cosmology Results And Evidence For Evolving Dark Energy With An Updated Type Ia Supernova Calibration
Porqueres, N., Enßlin, T. A., Greiner, M., et al. 2017, Astronomy & Astrophysics, 599, A92
Riess, A. G., Yuan, W., Macri, L. M., et al. 2022, The Astrophysical Journal Letters, 934, L7
Rubin, D., Aldering, G., Betoule, M., et al. 2023, Union Through UNITY: Cosmology with 2,000 SNe Using a Unified Bayesian Framework
Scolnic, D., Brout, D., Carr, A., et al. 2022, The Astrophysical Journal, 938, 113
Seikel, M., Clarkson, C., & Smith, M. 2012, JCAP, 06, 036
Shafieloo, A., Alam, U., Sahni, V., & Starobinsky, A. A. 2006, Monthly Notices of the Royal Astronomical Society, 366, 1081
Shafieloo, A., Kim, A. G., & Linder, E. V. 2012, Phys. Rev. D, 85, 123530
Sharma, R., Mukherjee, A., & Jassal, H. 2022, The European Physical Journal Plus, 137
Suzuki, N., Rubin, D., Lidman, C., et al. 2012, The Astrophysical Journal, 746, 85
Vincenzi, M., Brout, D., Armstrong, P., et al. 2024, ApJ, 975, 86
Vincenzi, M., Kessler, R., Shah, P., et al. 2025, Comparing the DES-SN5YR and Pantheon+ SN cosmology analyses: Investigation based on "Evolving Dark Energy or Supernovae systematics?"
Zhang, J.-C., Hu, Y., Jiao, K., et al. 2024, The Astrophysical Journal Supplement Series, 270, 23
Zheng, W. & Li, H. 2017, Astroparticle Physics, 86, 1

Appendix A: geoVI transform and standardization

Any variable $s \hookleftarrow \mathcal{G}(s, S)$ can be reparameterized in terms of a standard normal "latent variable" $\xi_s \hookleftarrow \mathcal{G}(\xi_s, \mathbf{1})$. This shifts the complexity of the correlation structure within the probability distribution to the mapping between s and ξ_s, allowing to construct hierarchical prior models (see Fig. 1) more easily. For a Gaussian field s, the mapping is achieved by considering an amplitude operator $A = F^\dagger \sqrt{\widehat{p_s}}$, where F is the Fourier transform, and $\widehat{p_s}$ is an operator with the signal power spectrum p_s as its diagonal: the field s can then be written as

$$s = A\xi_s, \tag{A.1}$$

which is a zero-mean, Gaussian variable with the covariance $\langle ss^\dagger\rangle_{(s)} = A\langle\xi_s\xi_s^\dagger\rangle_{(\xi_s)}A^\dagger = AA^\dagger = F^\dagger(\sqrt{\widehat{p_s}})^2F = S$. Note that the power spectrum p_s and by extension the amplitude operator A, are themselves built from standard normal variables (cf. Fig. 1). All standard normal variables involved in generating the model are collected into a single latent parameter vector ξ. To find an approximation to the posterior $p(\xi) := \mathcal{P}(\xi|d)$, the KL-divergence between the posterior and a parameterized trial distribution $\mathcal{Q}_\theta(\xi)$ is minimized. The KL-divergence reads

$$\mathcal{D}_{\mathrm{KL}}(\mathcal{Q}_\theta(\xi) \parallel p(\xi)) = \int \mathrm{D}_\xi\, \mathcal{Q}_\theta(\xi) \ln\left(\frac{\mathcal{Q}_\theta(\xi)}{p(\xi)}\right), \tag{A.2}$$

where D_ξ denotes an integration over all possible field configurations of ξ. In order to ensure that a Gaussian distribution, completely parameterized through a variable θ,

$$\mathcal{Q}_\theta(\xi) = \mathcal{G}(\theta, \Xi(\theta)),$$

is a good approximating distribution, geoVI performs a coordinate transformation that aims to flatten out potential curvature in the posterior distribution. To extract the necessary geometrical information from the posterior, a metric $\mathcal{M}$ is introduced:

$$\mathcal{M} = \mathrm{F}(\theta) + \mathbf{1}, \tag{A.3}$$

with the Fisher Information Metric

$$\mathrm{F}(\theta) = \left\langle \frac{\partial \mathcal{H}(d \mid \xi)}{\partial \xi} \frac{\partial \mathcal{H}(d \mid \xi)}{\partial \xi^\dagger} \right\rangle_{(d|\xi=\theta)}.$$

A coordinate transformation $g(\xi)$ is desired through which the metric, Eq. (A.3), becomes "diagonal" in the following sense:

$$\mathcal{M} \overset{!}{=} \left(\frac{\partial g}{\partial \xi}\right)^\dagger \mathbf{1} \left(\frac{\partial g}{\partial \xi}\right). \tag{A.4}$$

Frank et al. (2021) show that this transformation does not exist globally, but an expression that approximates the metric around an expansion $\bar{\xi}$ does:

$$g(\xi, \bar{\xi}) = \sqrt{\bar{\mathcal{M}}}^{-1}\left(\xi - \bar{\xi} + \left(\frac{\partial y}{\partial \xi}\bigg|_{\bar{\xi}}\right)^{\mathrm{T}} (y(\xi) - y(\bar{\xi}))\right) \tag{A.5}$$

where $\bar{\mathcal{M}}$ is the local metric at the expansion point and $y(\xi)$ is a function which through which the Fisher Information Metric also becomes diagonal:

$$\mathrm{F}(\theta) \overset{!}{=} \left(\frac{\partial y}{\partial \xi}\right)^\dagger \mathbf{1} \left(\frac{\partial y}{\partial \xi}\right)\bigg|_{\xi=\theta}. \tag{A.6}$$

For a Gaussian likelihood, the Fisher Information Metric can be calculated as

$$\mathrm{F}(\theta) = \left(\frac{\partial s'}{\partial \xi}\right)^\dagger N^{-1} \left(\frac{\partial s'}{\partial \xi}\right)\bigg|_{\xi=\theta}, \tag{A.7}$$

where s' is the noiseless data, $s' := R(s(\xi))$. Finally, a comparison between Eq. (A.6) and Eq. (A.7) suggests

$$y = \sqrt{N^{-1}}\, s',$$

where the noise covariance N is provided together with the distance moduli analyzed in this work. To take the matrix square root, we diagonalize the inverse noise covariance matrix and take the square root of the diagonal elements:

$$\sqrt{N^{-1}} = O\sqrt{\Lambda}O^\dagger, \tag{A.8}$$

where $\sqrt{\Lambda}$ is a diagonal matrix containing the square root of the eigenvalues of N^{-1} and O is the matrix consisting out of the unit-length eigenvectors of N^{-1}.

Appendix B: Construction of reference fields

Fig. 6 shows the signal for the cosmological parameters (cf. Table B.1) found by the Planck Collaboration (2020) (s_{CMB}) and Scolnic et al. (2022) (s_{SN}). These fields are parametric flat ΛCDM models, i.e.

$$\frac{H(a)^2}{H_0^2} = \Omega_\Lambda + \Omega_m a^{-3}. \tag{B.1}$$

This corresponds to the signal (see Eq. 8) of

$$s(x) = \ln\left(\frac{3}{8\pi G} H_0^2 (1 + \Omega_m (e^{3x} - 1))\right). \tag{B.2}$$

The comparison of s_{CMB} and s_{SN} illustrates the impact of different H_0 values on the signal's shape and amplitude. The 1σ regions in Fig. 6 are calculated through Gaussian propagation of the errors in Table B.1 following to Eq. (B.2).

Table B.1: Adopted values of H_0 and Ω_m from Pantheon+SH0ES and Planck2018 cosmological results, used for constructing flat ΛCDM reference fields to compare with charm1 and charm2 reconstructions.

	Pantheon+SH0ES Scolnic et al. (2022)	**Planck2018** Planck Collaboration (2020)
$\hat{H}_0$	73.60 ± 1.1	67.36 ± 0.54
Ω_m	0.334 ± 0.018	0.3153 ± 0.0073

Appendix C: Converting charm1 signal fields

For a meaningful comparison of the reconstructions between charm1 and charm2, we need to make adjustments due to slightly different signal definitions.

We translate the charm1 signal field, denoted by $s_{\mathtt{charm1,\,before}} := \ln(\rho/\rho_{c,0})$, to the corresponding charm2 signal field satisfying the new definitions Eq. (5) and Eq. (9) through

$$s_{\mathtt{charm1,\,after}} = \ln((\rho_{c,0}/\rho_0)e^{s_{\mathtt{charm1,\,before}}}), \tag{C.1}$$

where ρ_0 is the reference energy density introduced in this work, Eq. (9). The original charm1 reconstruction used only the Union2.1 dataset, which, at that time, was the latest available.

Appendix D: Details on the evidence lower bound

Writing $q(\xi)$ for the approximating distribution, $p(\xi)$ for the posterior and $\mathcal{H}_{\mathcal{P}}$ for the negative natural logarithm of probability $\mathcal{P}$, Eq. (A.2) can be rewritten as

$$\ln \mathcal{P}(d) = \mathcal{D}_{\mathrm{KL}}(q(\xi) \,\|\, p(\xi)) + \langle \mathcal{H}_q(\xi) \rangle_q - \langle \mathcal{H}_p(d, \xi) \rangle_q. \tag{D.1}$$

The ELBO is defined as

$$\mathrm{ELBO} := \langle \mathcal{H}_q(\xi) \rangle_q - \langle \mathcal{H}_p(d, \xi) \rangle_q. \tag{D.2}$$

The first term, the average of the approximating distribution's negative logarithm, can be calculated to be

$$\langle \mathcal{H}_q(\xi) \rangle_q = \frac{1}{2} \ln |2\pi e \Xi|, \tag{D.3}$$

where e is Euler's number and Ξ is the posterior covariance of q. The second term can be split into a likelihood and prior term:

$$\langle \mathcal{H}_p(d, \xi) \rangle_q = \langle \mathcal{H}_p(d|\xi) \rangle_q + \langle \mathcal{H}(\xi) \rangle_q. \tag{D.4}$$

Thanks to the standardization procedure, $\mathcal{P}(\xi)$ is by construction standard normal and the second term can be calculated as

$$\langle \mathcal{H}(\xi) \rangle_q = \frac{1}{2} \left(N \ln(2\pi) + \mathrm{Tr}(\Xi) + \bar{\xi}^\dagger \bar{\xi} \right), \tag{D.5}$$

where $\bar{\xi}$ denotes the posterior mean of q. Collecting all the terms and using the identity $\ln \det(\Xi) = \mathrm{Tr} \ln(\Xi)$, one finds

$$\begin{aligned} \mathrm{ELBO} &= -\langle \mathcal{H}(d|\xi) \rangle_q + \underbrace{\frac{1}{2} \mathrm{Tr} \ln(\Lambda_\Xi) - \frac{1}{2} \mathrm{Tr}(\Lambda_\Xi - \mathbf{1}) - \frac{1}{2} \bar{\xi}^\dagger \bar{\xi}}_{=:A} \\ &= -\langle \mathcal{H}(d|\xi) \rangle_q + A. \end{aligned} \tag{D.6}$$

Here, Λ_Ξ denotes the diagonalized matrix of the posterior covariance Ξ, needed for the calculation of the log determinant. Note that the ELBO decreases with an increase in the Occam's Razor term, $\frac{1}{2} \bar{\xi}^\dagger \bar{\xi}$: the more degrees of freedom need to deviate from their prior mean of 0 in latent space to explain the data, the less likely the model is.

Comparing Eq. (D.1) and Eq. (D.2), it is clear that the ELBO calculated this way provides a lower bound on the log-evidence:

$$\ln \mathcal{P}(d) \geq \mathrm{ELBO}. \tag{D.7}$$